



Indian Institute of Information Technology UNA Himachal Pradesh
An Institute of National Importance under MoE
Saloh, Una – 177209

Website: www.iiitu.ac.in

AY 2024-25

School of computing

CURRICULUM: IIITUGCSE22

Cycle Test – I

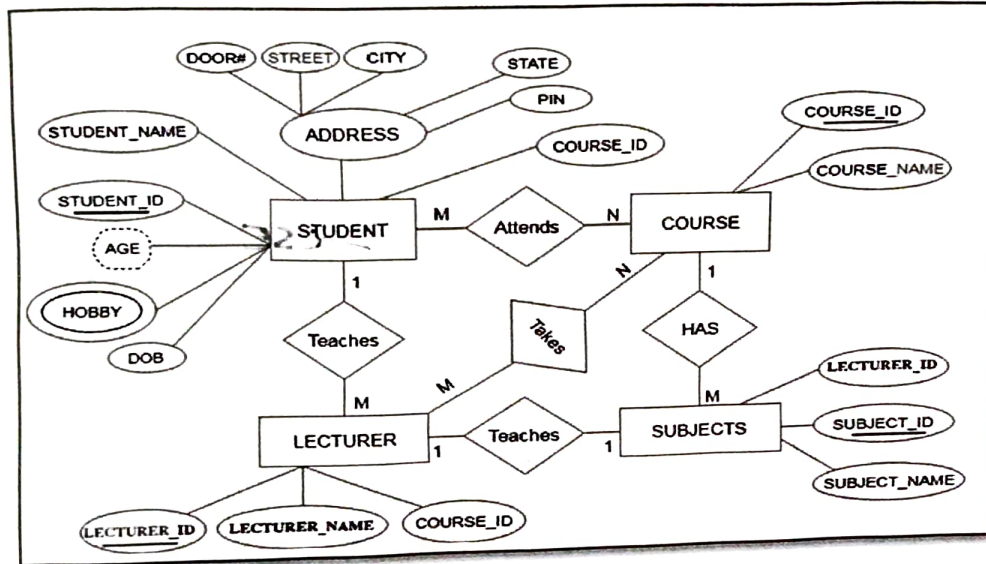
19, Feb.'25

Degree	B. Tech.	Branch	ITCSE
Semester	IV		
Subject Code & Name	CSC401 / DBMS		
Time: 60 Minutes (9:00-10:00 AM)	Answer All Questions	Maximum: 20 Marks	

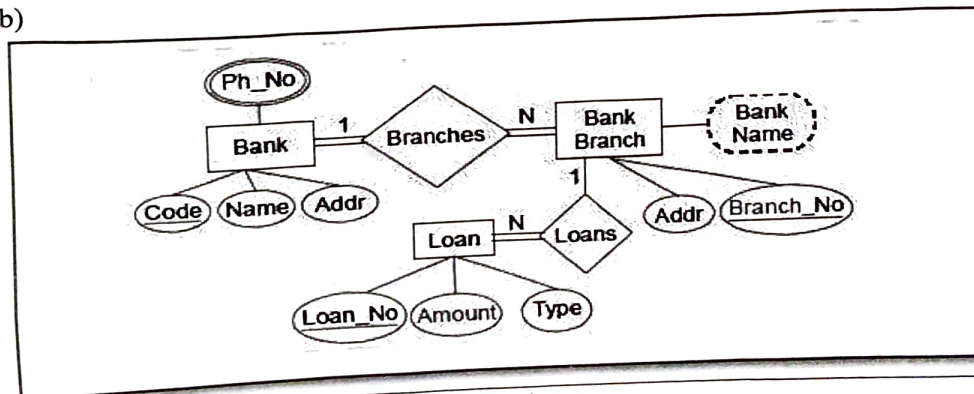
(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

Sl. No.	Question	Marks
1. a	Define Metadata, Intension and Extension of the database with example.	1
1. b	Find minimum number of tables required for converting the following entity relationship diagram into relational database. (a) Data, Info, DB, DBMS	2 (b) Data Ind types
1. c	What are the problems in file system data management? Explain in detail with relevant example.	2
2. a	Consider the relation scheme $R = \{E, F, G, H, I, J, K, L, M, N\}$ and the set of functional dependencies $\{ \{E, F\} \rightarrow \{G\}, \{F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\}, K \rightarrow \{M\}, L \rightarrow \{N\} \}$ on R. What is the key for R?	1
2. b	What are the different types of data model? Explain each briefly	2
2. c	Explain the Three Schema Architecture of a database with neat diagram	2
3. a	Consider a relation scheme $R = (A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$. What are the candidate keys of R?	1
3. b	Explain various Data Definition Commands in details with syntax & examples.	2
3. c	Define E/R Model. Explain the following: 3M a. Entities and Relationships 4M b. Attributes and different types of attributes in details with appropriate example.	2
4. a	We are given the relation $R(A, B, C, D, E)$. This means that the table R has five columns: A, B, C, D, and E. We are also given the set of functional dependencies: $\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E\}$. What is $\{A\}^+$?	1
4. b	Explain the differences between the following with example: a. Super key b. Candidate key c. Primary key d. Alternate Key	1

4. c Find a minimum number of tables along with the table name and their attributes required for converting the following entity relationship diagram into a relational database.



b)





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CURRICULUM: IIITUGCSE22

Cycle Test – II

15, April.'25

Degree	B. Tech.	Branch	CSE
Semester	IV		
Subject Code & Name	CSC401 / Database Management Systems		
Time: 60 Minutes	Answer All Questions	Maximum: 20 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

S. No.	Question	Marks																																	
1. a	The following functional dependencies hold true for the relational schema $R\{V,W,X,Y,Z\}$: $V \rightarrow W$ $VW \rightarrow X$ $Y \rightarrow VX$ $Y \rightarrow Z$ find the irreducible equivalent for this set of functional dependencies.	1																																	
1. b	Given below are two sets of FD's for a relation $R(A, B, C, D, E)$. Are they equivalent? $F=\{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G=\{A \rightarrow CD, E \rightarrow AH\}$	1																																	
1. c	i. Consider a relation $R(A,B,C,D,E)$ with the following three functional dependencies. $AB \rightarrow C; BC \rightarrow D; C \rightarrow E; AB \rightarrow C; BC \rightarrow D; C \rightarrow E;$ The number of super keys in the relation R is . ii. The maximum number of super keys for the relation schema $R(E, F, G, H)$ with E as the key is -----.	2																																	
2. a	i. What is a Normalization? Explain the 1NF, 2NF & 3NF with examples. ii. What is Functional dependency? Explain the inference rules for functional dependency with proof.	3																																	
2. b	Consider the following three schedules of transactions T_1, T_2 and T_3 . [Notation: In the following NYO represents the action Y (R for read, W for write) performed by transaction N on object O .] <table><tr><td>(S1)</td><td>2RA</td><td>2WA</td><td>3RC</td><td>2WB</td><td>3WA</td><td>3WC</td><td>1RA</td><td>1RB</td><td>1WA</td><td>1WB</td></tr><tr><td>(S2)</td><td>3RC</td><td>2RA</td><td>2WA</td><td>2WB</td><td>3WA</td><td>1RA</td><td>1RB</td><td>1WA</td><td>1WB</td><td>3WC</td></tr><tr><td>(S3)</td><td>2RA</td><td>3RC</td><td>3WA</td><td>2WA</td><td>2WB</td><td>3WC</td><td>1RA</td><td>1RB</td><td>1WA</td><td>1WB</td></tr></table> Which of the above schedules are conflict-serializable also find the order of execution of schedule?	(S1)	2RA	2WA	3RC	2WB	3WA	3WC	1RA	1RB	1WA	1WB	(S2)	3RC	2RA	2WA	2WB	3WA	1RA	1RB	1WA	1WB	3WC	(S3)	2RA	3RC	3WA	2WA	2WB	3WC	1RA	1RB	1WA	1WB	3
(S1)	2RA	2WA	3RC	2WB	3WA	3WC	1RA	1RB	1WA	1WB																									
(S2)	3RC	2RA	2WA	2WB	3WA	1RA	1RB	1WA	1WB	3WC																									
(S3)	2RA	3RC	3WA	2WA	2WB	3WC	1RA	1RB	1WA	1WB																									
3 a	Let $R(A,B,C,D)$ be a relational schema in which the following functional dependencies are known to hold:	1																																	

$A \rightarrow B$
 ~~$B \rightarrow A$~~

The decomposition of R into $(A,B), (B,C), (B,D), (A,B)$ find whether the decomposition is lossless or lossy. Also find the whether the FD is dependency preserving or not?

3. b i. Consider the following log sequence of two transactions on a bank account, with initial balance 12000, that transfer 2000 to a mortgage payment and then apply a 5% interest.

- 1.T1 start
- 2.T1 B old =1200 new =10000
- 3.T1 M old =0 new =2000
- 4.T1 commit
- 5.T2 start
- 6.T2 B old =10000 new =10500
- 7.T2 commit

Suppose the database system crashes just before log record 7 is written. When the system is restarted, which statement is true of the recovery procedure and explain with proper justification

- We must redo log record 6.

- We must redo log record 6 to set B to 10500
 - We must undo log record 6 to set B to 10000 and then redo log records 2 and 3
 - We need not redo log records 2 and 3 because transaction T1 has committed
 - We can apply redo and undo operations in arbitrary order because they are idempotent
- ii. Consider the following tables T1 and T2.

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

R	S
2	2
8	3
3	2
9	7
5	7
7	2

In table T1, P is the primary key and Q is the foreign key referencing R in table T2 with ondelete cascade and on-update cascade. In table T2, R is the primary key and S is the foreign key referencing P in table T1 on-delete set NULL and on-update cascade. In order to delete record (3,8) from table T1, the number of additional records that need to be deleted from table T1 is -----.

4 a	i. A database of research articles in a journal uses the following schema. (VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE) The primary key is (VOLUME, NUMBER, STARTPAGE, ENDPAGE) and the following functional dependencies exist in the schema. (VOLUME, NUMBER, STARTPAGE, ENDPAGE) $\twoheadrightarrow$ TITLE (VOLUME, NUMBER) $\rightarrow$ YEAR (VOLUME, NUMBER, STARTPAGE, ENDPAGE) $\rightarrow$ PRICE The database is redesigned to use the following schemas. (VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, PRICE) (VOLUME, NUMBER, YEAR) Find the weakest normal form that the new database satisfies but the old one does not.	
4 b	ii. Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values. $F \rightarrow \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R. (a) Find how many Candidate keys does the relation R have? (b) Find the relation R is in which normal form? (c) What are the prime and non-prime attributes?	
	Define transaction. Discuss the ACID properties. neat and complete answer	



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CURRICULUM: IIITUGCSE22

End Semester Examination

05, May '25

Degree	B. Tech.	Branch	CSE/IT
Semester	IV		
Subject Code & Name	CSC401 ITC1404/ Database Management Systems		
Time: 3 hrs	Answer All Questions	Maximum: 100 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating, K6: Create)

S. No.	Question	Marks																														
1.a	i. Explain the three-level schema architecture with a neat diagram ii. What is data independence, and what are the types of data independence	(3,2) 5																														
1. b	i. Design a suitable schema for the employee database. Identify the attributes, entities, tuples, relations, degree, cardinality, and primary keys in it. ii. Discuss the Entity Relationship Model, its purpose, and the notations used in the ER model	5																														
1. c	i. Given the STUDENTS relation as shown below. <table border="1"><thead><tr><th>StudentID</th><th>StudentName</th><th>StudentEmail</th><th>StudentAge</th><th>CPI</th></tr></thead><tbody><tr><td>2345</td><td>Shankar</td><td>shankar@math</td><td>X</td><td>9.4</td></tr><tr><td>1287</td><td>Swati</td><td>swati@ee</td><td>19</td><td>9.5</td></tr><tr><td>7853</td><td>Shankar</td><td>shankar@cse</td><td>19</td><td>9.4</td></tr><tr><td>9876</td><td>Swati</td><td>swati@mech</td><td>18</td><td>9.3</td></tr><tr><td>8765</td><td>Ganesh</td><td>ganesh@civil</td><td>19</td><td>8.7</td></tr></tbody></table> For (StudentName, StudentAge) to be the key for this instance, the value X should not be equal to---- and also explain the reason behind the answer. ii. The relation scheme Student Performance (name, courseNo, rollNo, grade) has the following functional dependencies: name, courseNo $\rightarrow$ grade rollNo, courseNo $\rightarrow$ grade name $\rightarrow$ rollNo rollNo $\rightarrow$ name The highest normal form of this relation scheme is-----	StudentID	StudentName	StudentEmail	StudentAge	CPI	2345	Shankar	shankar@math	X	9.4	1287	Swati	swati@ee	19	9.5	7853	Shankar	shankar@cse	19	9.4	9876	Swati	swati@mech	18	9.3	8765	Ganesh	ganesh@civil	19	8.7	5
StudentID	StudentName	StudentEmail	StudentAge	CPI																												
2345	Shankar	shankar@math	X	9.4																												
1287	Swati	swati@ee	19	9.5																												
7853	Shankar	shankar@cse	19	9.4																												
9876	Swati	swati@mech	18	9.3																												
8765	Ganesh	ganesh@civil	19	8.7																												
1.d	i. Define Keys (1 mark)? ii. Explain super key, candidate key, primary, and alternate key with examples?	5																														
2. a	Consider a relation R (A, B, C, D, E) with three functional dependencies. i. $AB \rightarrow C$; $BC \rightarrow D$; $C \rightarrow E$. Suppose A is the candidate key, then the number of superkeys in the relation R is ----- ii. Suppose AB and AC are the candidate keys; how many superkeys are in the relation R? iii. Let R (A, B, C, D, E, F) be a relation scheme with the following dependencies: $C \rightarrow F$, $E \rightarrow A$, $EC \rightarrow D$, $A \rightarrow B$. Find out the candidate Key of R? iv. The following functional dependencies are given: $AB \rightarrow CD$, $AF \rightarrow D$, $DE \rightarrow F$, $C \rightarrow G$, $F \rightarrow E$, $G \rightarrow A$. Find the closure of AF^+	5																														

2. b	i. What do you mean by Normalisation? Explain with an example. Explain BCNF, 3NF, 2NF and 1NF with a suitable example. ii. Write four differences between the DROP, Truncate, and DELETE commands with syntax?	5
2. c	Consider a database of fixed-length records, stored as an ordered file. The database has 30,000 records, with each record being 100 bytes, of which the primary key occupies 15 bytes. The data file is block-aligned in that each data record is fully contained within a block. The database is indexed by a primary index file, which is also stored as a block-aligned ordered file. Suppose the block size of the file system is 1024 bytes, and a pointer to a block occupies 5 bytes. The system uses binary search on the index file to search for a record with a given key. You may assume that a binary search is on an index file. i. Find the blocking factor with and without primary indexing ii. Find the number of blocks required for the main and index files in the case of primary indexing. iii. Number of block accesses in the main and index files in the case of primary indexing. iv. Find the blocking factor with and without secondary indexing v. Find the number of blocks required for the main and index files in the case of secondary indexing. vi. Number of block accesses in the main and index files in the case of secondary indexing.	6
2. d	What is indexing? Why do we go for indexing? and what are the different types of indexing?	5
3. a	A data file consisting of 1,50,000 student records is stored on a hard disk with a block size of 4096 bytes. The data file is sorted using the primary key RollNo. The size of a record pointer for this disk is 7 bytes. Each student record has a candidate key attribute called A Num of size 12 bytes. Suppose an index file with records consisting of two fields, a Num value and the record pointer to the corresponding student record, is built and stored on the same disk. Assume that the records of the data file and the index file are not split across disk blocks. The number of blocks in the index file is	5
3. b	The order of a leaf node in a B+B+ - tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is 1K bytes, the data record pointer is 7 bytes long, the value field is 9 bytes long, and the block pointer is 6 bytes long, what is the order of the leaf node?	5
3. c	Why concurrency control in DBMS? Explain it in terms of the locking mechanism of the two-phase locking protocol.	5
3. d	Define data definition language. List various DDL and DML commands. Write the syntax for each command and explain them with suitable examples.	5
4. a	i. Write the difference between B-tree and B*tree with an example (write at least five differences) ii. Define log? Write four differences between immediate and deferred log-based recovery.	5
4. b	I. Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes. D1: R=[(P,QS,T);(P,T,X);(Q,Y);(Y,Z,W)] D2: R=[(P,Q,S);(T,X);(Q,Y);(Y,Z,W)] i. Find which one of the above decompositions, i.e. D1 or D2, is lossless or lossy. ii. Find whether D1 and D2 is dependency preserving or not. II. From the following instance of a relation scheme R (A, B, C), based on the functional dependency rules, which of the functional dependencies holds good from the below given relation	5

10
3000
3000
51
58
11

1024 = 10
100 = 3000
30000 = 3000
10
109 3000 = 12
1024 = 51
20
3000 = 51
51
109 51 = 7

512 (1024) 512
1024 ≤ 1024

A	B	C
1	1	1
1	1	0
2	3	2
2	3	2

~~A-B~~ B-A
~~A-C~~ C-A
~~B-C~~ C-B
~~A-B~~ BC-A
~~B-A~~ AC-B
~~A-C~~ C-AB

- 4.c Consider the transactions T1, T2, and T3 and the schedules S1 and S2 given below.
 T1 : r1(x); r1(Z); w1(x); w1(Z)
 T2 : r2(x); r2(Z); w2(Z)
 T3 : r3(x); r3(x); w3(Y)
 S1: r1(x); r3(Y); r3(x); r2(Y); r2(Z); w3(Y); w2(Z); r1(Z); w1(x); w1(Z)
 S2: r1(x); r3(Y); r2(Y); r3(x); r1(Z); r2(Z); w3(Y); w1(x); w2(Z); w1(Z)
 Which one of the following above schedules is Conflict serializable using the precedence graph, and find the view serializability?

5

- 4.d Consider schedules S1, S2, and S3 below. Determine whether each schedule is strict, cascadeless, recoverable, or non-recoverable. (Determine the strictest recoverability condition that each schedule satisfies.)
 S1: r1 (X); r2 (Z); r1 (Z); r3 (X); r3 (Y); w1 (X); c1; w3 (Y); c3; r2 (Y); w2 (Z); w2 (Y); c2;
 S2: r1 (X); r2 (Z); r1 (Z); r3 (X); r3 (Y); w1 (X); w3 (Y); r2 (Y); w2 (Z); w2 (Y); c1; c2; c3;
 S3: r1 (X); r2 (Z); r3 (X); r1 (Z); r2 (Y); r3 (Y); w1 (X); c1; w2 (Z); w3 (Y); w2 (Y); c3; c2;

5

5. a Given the following STUDENT COUSE scheme: STUDENT (Rollno, Name, Courseno), COURSE (courseno, coursename, capacity), where Rollno is the primary key of relation STUDENT and Courseno is the primary key of relation COURSE. Attribute Coursename of COURSE takes unique values only. The number of records in the COURSE and STUDENT tables is 3 and 5, respectively. Following relational algebra is executed. **R=STUDENT X COURSE**

5

- Find the degree of the table R and explain how.
- Find the cardinality of the table R and explain how.
- Foreign key of relation STUDENT and explain how.
- Foreign key of relation COURSE and explain how.

5. b Consider the following relations X (S, Si, C) and Y (S, P, D).

7

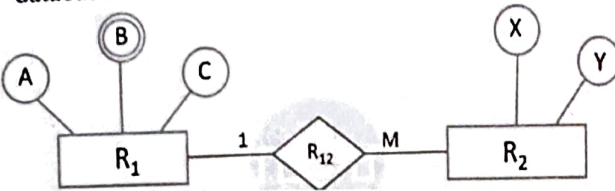
X

S	Si	C
J	1	M
B	2	N
R	3	H
T	4	G

Y

S	P	D
J	S ₁	CA
B	P ₁	AB
R	D ₁	DC
A	H ₁	MD

- $X \bowtie_{X.S=Y.S} Y$ is :
- Find the Result of the and also tell the number of tuples after applying the operation.
 - Find the number of tuples by applying the Cartesian product over X and Y. Write the head of 5 tuples. (5 rows)

	iii. Number of tuples by applying right outer join on relation X and Y is/are. Also, find the result in the form of rows, which are extracted after applying operations. iv. Number of tuples by applying left outer join on relation X and Y is/are. Also, find the result in rows extracted after applying operations. v. Number of tuples by applying full outer join on relation X and Y is/are. Also, find the result in the rows extracted after applying operations.-	
5. c	i. Find the minimum number of tables and also write the names of the tables with their attributes required for converting the following entity relationship diagram into a relational database?  ii. Let E1 and E2 be two entities in an E/R diagram with simple single-valued attributes. R1 and R2 are two relationships between E1 and E2, where R1 is one-to-many and R2 is many-to-many. R1 and R2 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model? iii. Given two tables R1 (X, Y) and R2 (Y, Z) with 50 and 30 number of tuples respectively. Find a maximum number of tuples in the output of the natural join between tables R1 and R2. Also, find the maximum number of tuples in the output of the Cartesian join (cross join) between tables R1 and R2.	4
5.d	An organization needs to maintain database having five attributes A, B, C, D, E. These attributes are functionally dependent on each other for which functional dependency set F is given as : F: $\{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$. Consider a universal relation R(A, B, C, D, E) with functional dependency set F. Also all attributes are simple and take atomic values only. i. Minimal Cover F' of functional dependency set F is: ii. Identify the redundant functional dependency in F iii. Identify the Primary key of table R with functional dependency set F iv. Identify the normal form in which relation R belong to:---	4

***** GOOD LUCK *****



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Himachal Pradesh

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School of Computing

CURRICULUM: IITUGIT22

END SEMESTER EXAMINATION

Date of Examination: 06/05/2024 (09:00 AM - 12:00 Noon)

Degree	B. Tech.	Branch	Information Technology
Semester	IV		
Subject Code & Name	ITC404 – Database Management Systems		
Time: 180 Minutes	Answer All Questions		Maximum: 100 Marks

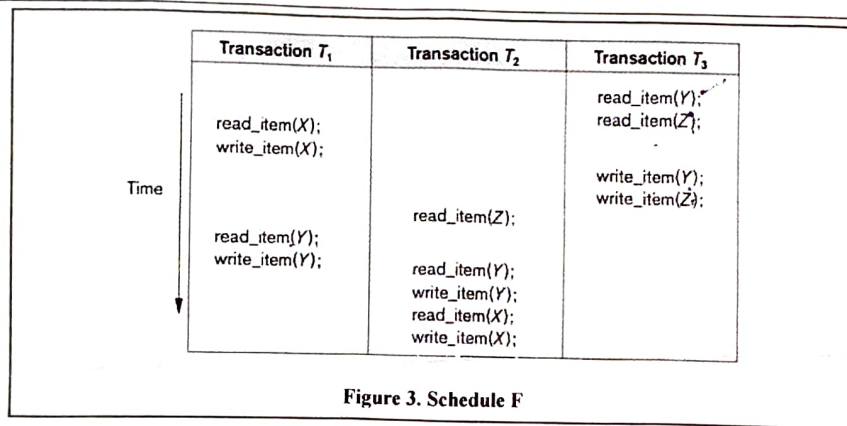
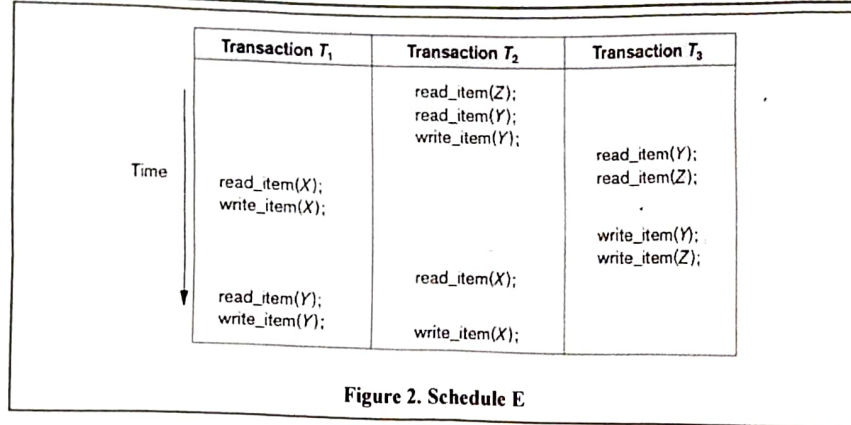
S. No.	Question	Marks
1.a	Figure 1. ER- diagram for a BANK database schema. Consider the ER diagram as shown in the Figure 1 as a part of BANK database. Each bank can have multiple branches, and each branch can have multiple accounts and loans. - List all the strong entity types in the ER diagram. - Is there a weak entity type? If so, give its name, partial key, and its identifying relationship.	(5)
1.b	What concepts do high-level or conceptual data models use? How are they different from low-level or physical data models.	(5)
1.c	How DBMS is classified on the basis of number of users supported by the database system?	(5)

	Differentiate it with the DBMS classification on the basis of number of sites over which the database is distributed. Give examples of each.																																																
1.d	Elaborate upon five distinct advantages of using the DBMS approach, which good DBMS should possess other than self-describing nature of database, insulation between programs and data, support of multiple views of the data, and multiple transaction processing.	(5)																																															
2.a	Consider the Table 1 and Table 2 as follows: Table 1. EMPLOYEE Table <table><tr><th>Fname</th><th>Bdate</th><th>Salary</th><th>SSn</th><th>Dno</th></tr><tr><td>A1</td><td>20-08-2002</td><td>30000</td><td>11111</td><td>3</td></tr><tr><td>B1</td><td>20-08-2002</td><td>50000</td><td>22222</td><td>3</td></tr><tr><td>C1</td><td>20-08-2002</td><td>20000</td><td>33333</td><td>2</td></tr><tr><td>D1</td><td>20-08-2002</td><td>25000</td><td>44444</td><td>1</td></tr><tr><td>E1</td><td>20-08-2002</td><td>18000</td><td>55555</td><td>2</td></tr><tr><td>F1</td><td>20-08-2002</td><td>40000</td><td>66666</td><td>1</td></tr></table> Table 2. DEPARTMENT Table <table><tr><th>Dname</th><th>Dnumber</th><th>MSn</th></tr><tr><td>D11</td><td>3</td><td>33333</td></tr><tr><td>D22</td><td>1</td><td>11111</td></tr><tr><td>D33</td><td>2</td><td>66666</td></tr></table> Determine the output of the following operations:- i. $\sigma_{(Dno=1 \text{ AND } Salary>25000) \text{ OR } (Dno=3 \text{ AND } Salary>30000)}(EMPLOYEE)$ ii. $\pi_{Bdate, Fname}(EMPLOYEE)$ iii. Suppose R1 is the result of 2.a (i) and R2 is the result of 2.a (ii), what will be $R1 \times R2$.	Fname	Bdate	Salary	SSn	Dno	A1	20-08-2002	30000	11111	3	B1	20-08-2002	50000	22222	3	C1	20-08-2002	20000	33333	2	D1	20-08-2002	25000	44444	1	E1	20-08-2002	18000	55555	2	F1	20-08-2002	40000	66666	1	Dname	Dnumber	MSn	D11	3	33333	D22	1	11111	D33	2	66666	(1+1+3)
Fname	Bdate	Salary	SSn	Dno																																													
A1	20-08-2002	30000	11111	3																																													
B1	20-08-2002	50000	22222	3																																													
C1	20-08-2002	20000	33333	2																																													
D1	20-08-2002	25000	44444	1																																													
E1	20-08-2002	18000	55555	2																																													
F1	20-08-2002	40000	66666	1																																													
Dname	Dnumber	MSn																																															
D11	3	33333																																															
D22	1	11111																																															
D33	2	66666																																															
2.b	Given two union-compatible relations R and S comprising of at least three different tuples. Define the three operations Union, Intersection and Set Difference on them.	(5)																																															
2.c	Show how can one specify the following relational algebra operations in both tuple and domain relational calculus: i. $R(A, B) \div S(A)$ ii. $R(A, B, C) * S(C, D, E)$	(2.5 + 2.5)																																															
2.d	What are the DDL and DML commands in SQL? Give appropriate examples of each. Identify the key difference in using DELETE, DROP and TRUNCATE keywords in SQL.	(5)																																															
3.a	How can the key and foreign key constraints be enforced by the DBMS? Is the enforcement technique one suggests difficult to implement? Can the constraint checks be executed efficiently when updates are applied to the database? Justify.	(5)																																															
3.b	Prove that any relation schema with two attributes is in BCNF.	(5)																																															
3.c	Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I\}$ and the set of functional dependencies $F = \{ \{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\} \}$. What is the key for R? Decompose R into 2NFrelations.	(5)																																															
3.d	Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I\}$ and the set of functional dependencies $F = G = \{ \{A, B\} \rightarrow \{C\}, \{B, D\} \rightarrow \{E, F\}, \{A, D\} \rightarrow \{G, H\}, \{A\} \rightarrow \{I\}, \{H\} \rightarrow \{J\} \}$. List the candidate keys for R. Determine the 3NF relations for R.	(5)																																															
4.a	What are the characteristic features of the Log Buffer, which is used to recover from failures that affect transactions? What are the types of entries involved in it? Explain each of them using	(5)																																															

appropriate examples.

4.b

(5)



Perform serializability tests for the given Schedules E and F in Figure 2 and Figure 3. Identify it using the precedence graphs. Determine equivalent serial schedule if any.

4.c

Which of the following schedules is conflict serializable? For each serializable schedule, determine the equivalent serial schedules.

(5)

- i. $r_1(X); r_3(X); w_1(X); r_2(X); w_3(X);$
- ii. $r_1(X); r_3(X); w_3(X); w_1(X); r_2(X);$
- iii. $r_3(X); r_2(X); w_3(X); r_1(X); w_1(X);$
- iv. $r_3(X); r_2(X); r_1(X); w_3(X); w_1(X);$

4.d

Consider schedules S1, and S2 below. Determine whether each schedule is strict, cascadeless, recoverable, or non-recoverable. Justify the reasons for each separately.

(2.5 +
2.5
)

S1: $r_1(x); r_2(z); r_1(z); r_3(x); r_3(y); w_1(x); c_1; w_3(y); c_3; r_2(y); w_2(z); w_2(y); c_2$

S2: $r_1(x); r_2(z); r_3(x); r_1(z); r_2(y); r_3(y); w_1(x); w_2(z); w_3(y); w_2(y); c_3; c_2;$

5.a

Consider a disk with the following characteristics: block size $B=512$ bytes, interblock gap size $G=128$ bytes, number of blocks per track=20, number of tracks per surface=400. A disk pack consists of 15 doublesided disks.

(1×5)

- i. What is the total capacity of a track and what is its useful capacity (excluding interblock gaps)?
- ii. How many cylinders are there?
- iii. What is the total capacity and the useful capacity of a cylinder?
- iv. What is the total capacity and the useful capacity of a disk pack?
- v. Suppose the average seek time is 30 msec. How much time does it take (on

	the average) in msec to locate and transfer a single block given its block address?	
5.b	Draw a neat and clean diagram showing a disk pack. Define the following terms in reference to it: i. Track. ii. Cylinder iii. Sectors iv. Blocks	(1+1×4)
5.c	How does multilevel indexing improve the efficiency of searching an index file? How does a B-tree differ from a B ⁺ -tree? Why is a B ⁺ tree usually preferred as an access structure to a data file?	(2+1+2)
5.d	Describe External Hashing in detail. Which collision resolution methods are used for implementing Hashing? Differentiate between Dynamic and Extendible Hashing.	(2+2+1)
** ALL THE BEST **		



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School of Computing CURRICULUM: IIITUGIT22

Cycle Test – I

19 February 2024

Degree	B. Tech.	Branch	IT
Semester	IV		
Subject Code & Name	ITC404 : Database Management Systems		
Time: 60 Minutes	Answer All Questions	Maximum: 20 Marks	

Sl. No.	Question	Marks																														
1.a	Propose an appropriate architecture for database systems. Differentiate between a database schema and database state.	(1)																														
1.b	Identify four distinct characteristics of the database approach and specify how do they differ from the traditional file systems.	(2)																														
1.c	Design a suitable relational schema for the HR department in a company. Identify the Attributes, Tuples and Relations in it uniquely, taking appropriate examples.	(2)																														
2.a	Show how you can specify the given relational algebra operations in tuple and domain relational calculus: $R(A, B, C) \times S(D, E, F)$	(1)																														
2.b	Figure 1. TOYS table <table><thead><tr><th>toy_id</th><th>toy_name</th><th>toy_colour</th></tr><tr><th>integer</th><th>character varying (20)</th><th>character varying (10)</th></tr></thead><tbody><tr><td>1</td><td>Miss Shoppies</td><td>pink</td></tr><tr><td>2</td><td>Cuteasaurus</td><td>blue</td></tr><tr><td>3</td><td>Baby Turtle</td><td>green</td></tr></tbody></table> Figure 2. BRICKS table <table><thead><tr><th>brick_id</th><th>brick_colour</th><th>brick_shape</th></tr><tr><th>integer</th><th>character varying (10)</th><th>character varying (10)</th></tr></thead><tbody><tr><td>1</td><td>blue</td><td>cube</td></tr><tr><td>2</td><td>green</td><td>cube</td></tr><tr><td>3</td><td>blue</td><td>pyramid</td></tr></tbody></table> TOYS table is given as in Figure 1 and BRICKS table is given as in Figure 2. Determine the output for the following query: <code>select toy_id from TOYS UNION select brick_id from BRICKS;</code> <code>select * from toys inner join bricks on toy_id <= brick_id;</code>	toy_id	toy_name	toy_colour	integer	character varying (20)	character varying (10)	1	Miss Shoppies	pink	2	Cuteasaurus	blue	3	Baby Turtle	green	brick_id	brick_colour	brick_shape	integer	character varying (10)	character varying (10)	1	blue	cube	2	green	cube	3	blue	pyramid	(2)
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integer	character varying (10)	character varying (10)																														
1	blue	cube																														
2	green	cube																														
3	blue	pyramid																														
2.c	Define a primary key. How is it different to a superkey or a candidate key? Illustrate, using appropriate examples.	(2)																														
3.a	Given two union compatible relations R and S, define the two operations SET INTERSECTION and SET DIFFERENCE. How are these related?	(1)																														

3.b

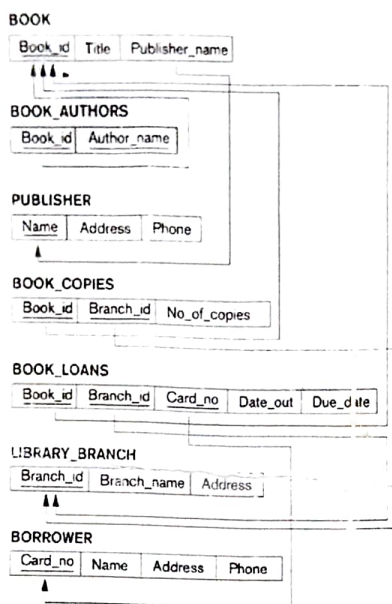


Figure 3. LIBRARY database schema

(2)

Considering the given schema as shown in Figure 3, which is used to keep the track of books, their borrowers and book loans. The referential integrity constraints are shown using the directed arcs. Derive the relational expressions for the following queries:

- For each library branch, retrieve the branch name and the total number of books loaned out from that branch.
- How many copies of the book having a title comprising of letter 'a' twice (may not be successive) in the BOOK table are owned by a library branch whose name is 'XYZ'.

3.c

Given the following instances of tuples in the relation, determine which of the given functional dependencies hold in it. Justify the answer.

RollNo	Name
11	Lakshay
11	Prince
12	Prince

(2)

Functional Dependency set $FD = \{ RollNo \rightarrow Name, Name \rightarrow RollNo \}$

4.a

Given a relation $R(P, Q, R, S, T)$ and Functional Dependency set $FD = \{ PQ \rightarrow R, S \rightarrow T \}$, determine whether the given R is in 2NF using normalization theory rules. If not, convert it into 2NF.

(1)

4.b

Why should NULLs in a relation be avoided as much as possible? Discuss the problem of spurious tuples with examples and how one may prevent it.

(2)

4.c

Illustrate the types of possible update anomalies using appropriate examples. Why are they considered bad?

(2)